

15. Implement probabilistic context-free grammar parsing for a sentence using python

Python Code

```
grammar = {
    ("S", "NP VP"): 1.0,
    ("NP", "John"): 0.6,
    ("NP", "Mary"): 0.4,
    ("VP", "runs"): 0.7,
    ("VP", "eats"): 0.3
}

sentence = input("Enter sentence: ").split()

if len(sentence) == 2:
    subject, verb = sentence
    if ("NP", subject) in grammar and ("VP", verb) in grammar:
        probability = grammar[("S", "NP VP")] * \
            grammar[("NP", subject)] * \
            grammar[("VP", verb)]
        print("Sentence Accepted")
        print("Probability =", probability)
    else:
        print("Sentence Rejected")
else:
    print("Sentence Rejected")
```

Input

John runs

Output

Sentence Accepted

Probability = 0.42

The image shows a Windows 11 desktop environment. On the left, a Jupyter Notebook is open in a web browser. The code in the notebook defines a function to parse a sentence into subject and verb using a simple grammar. The input sentence is "John runs", and the output shows the probability of 0.42 for the sentence being accepted. On the right, the Gemini AI chat interface is open. It displays a greeting "Hello, SRAVYA" and offers to help with tasks like explaining POS tags or performing Named Entity Recognition. The chat history shows a previous interaction where the user asked how to filter and display only nouns from POS tags. The bottom of the screen shows the Windows taskbar with various application icons and the system tray displaying the time as 14:00 on 28-07-2023.